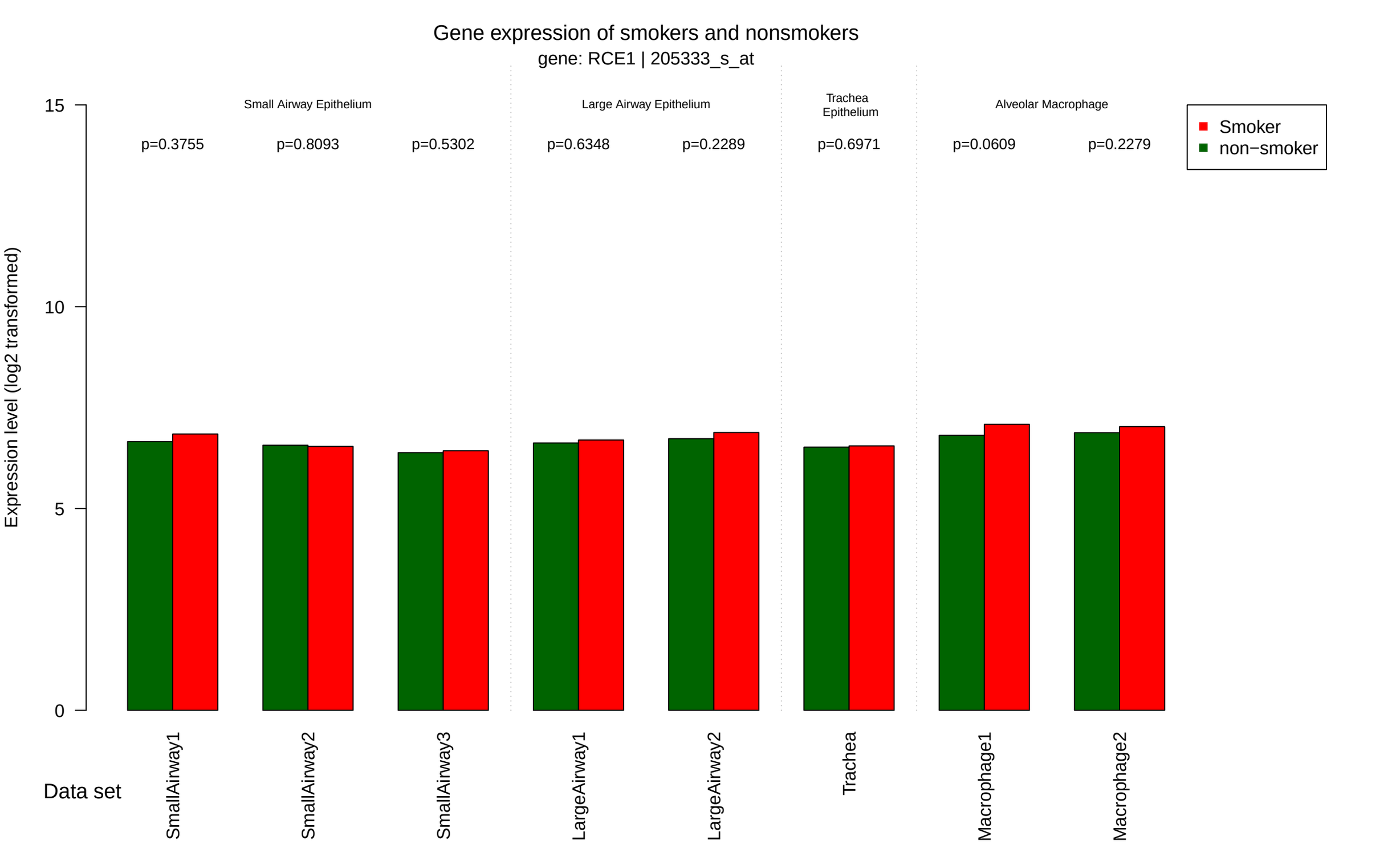
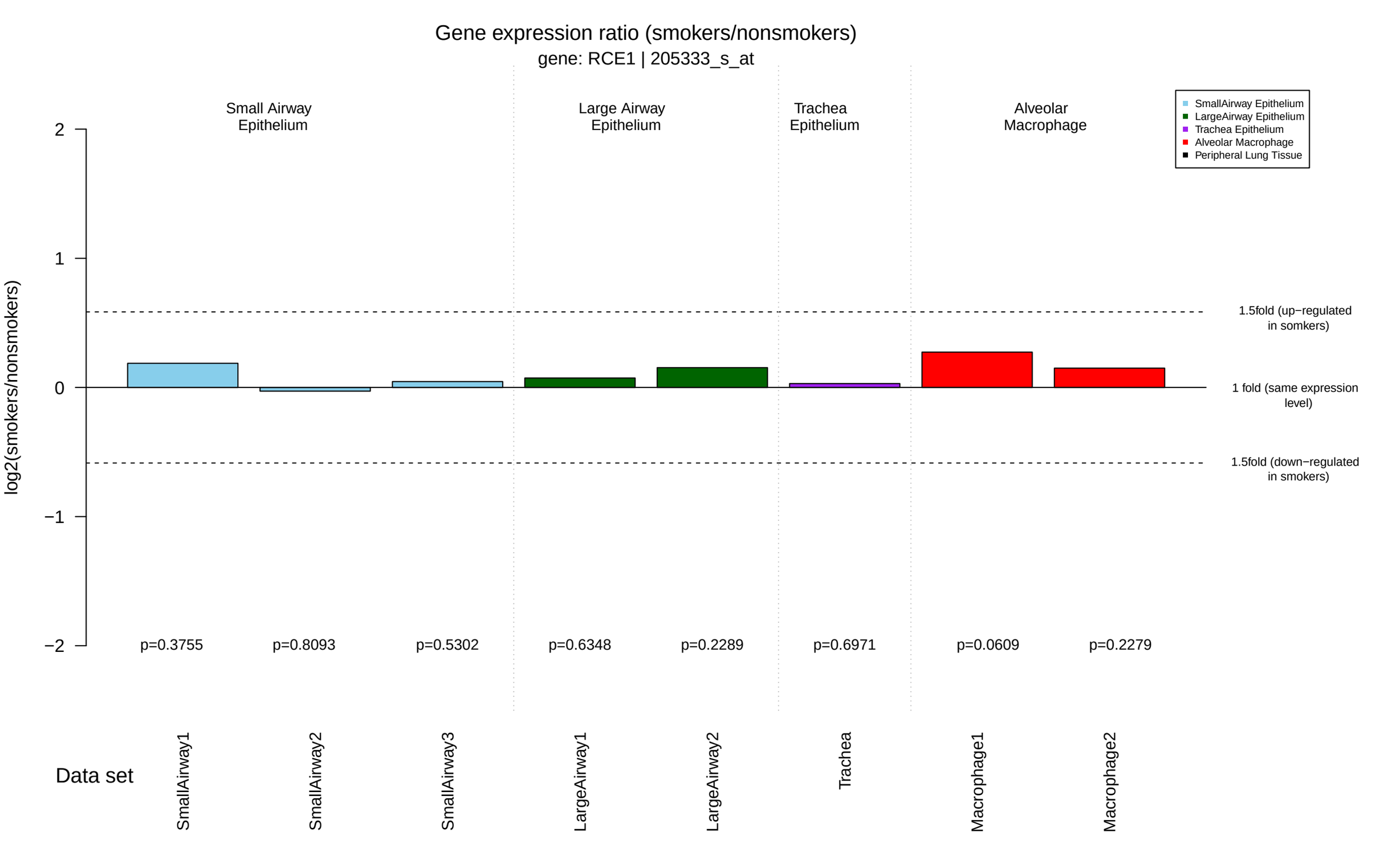
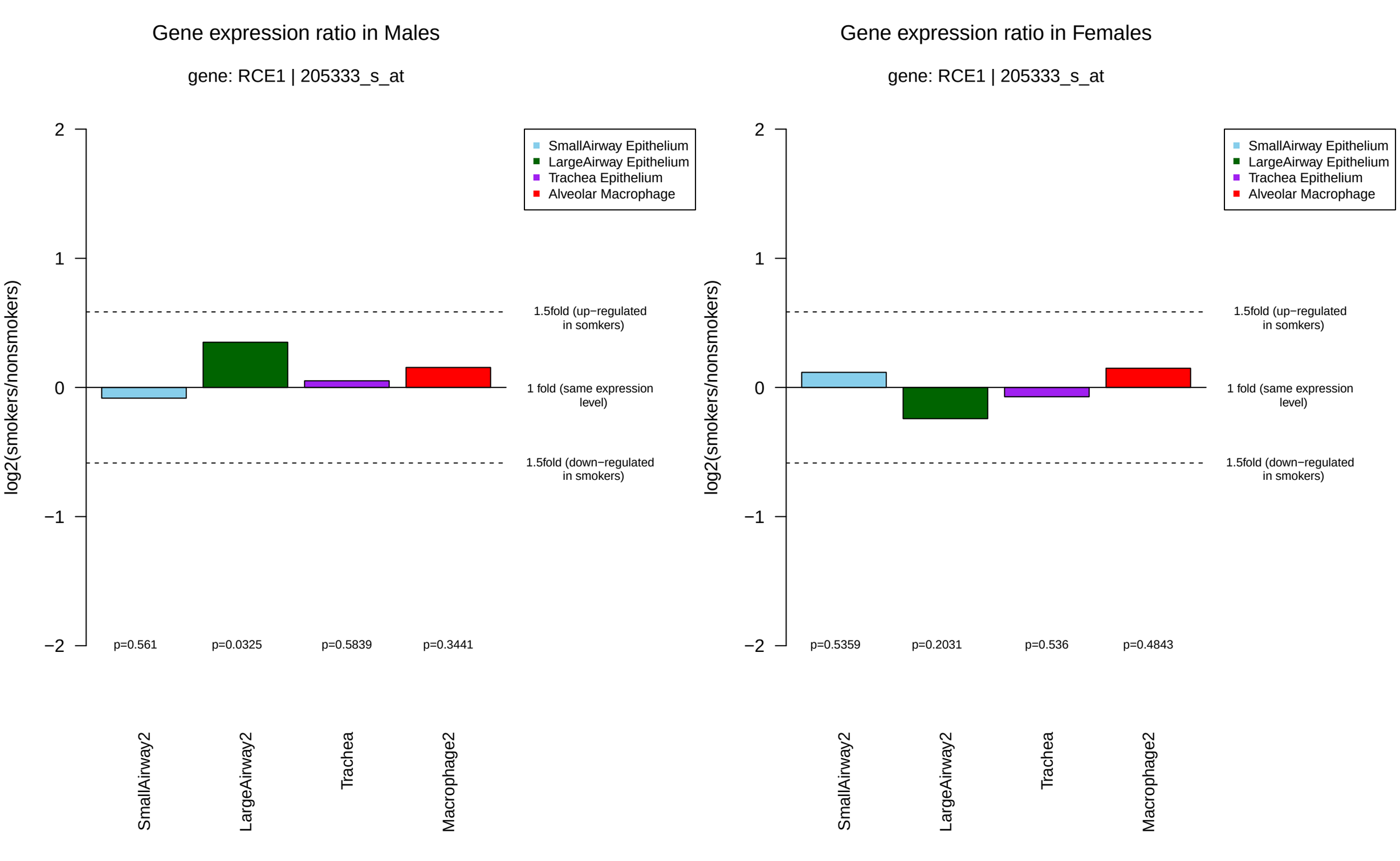
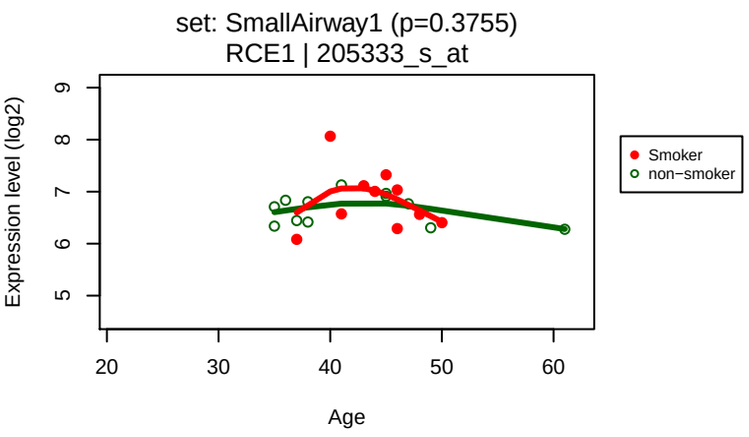
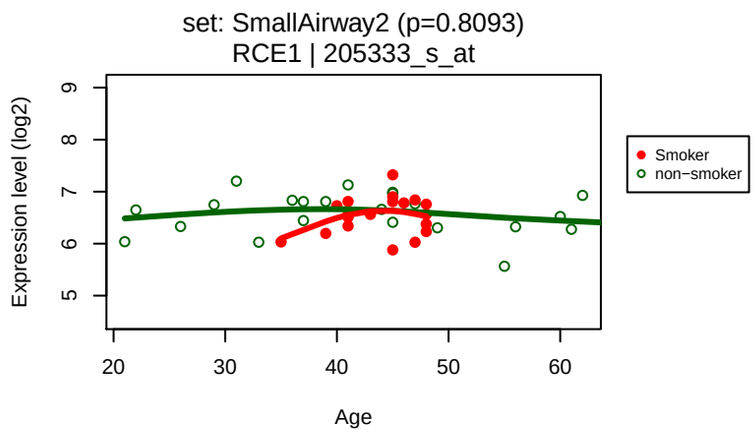
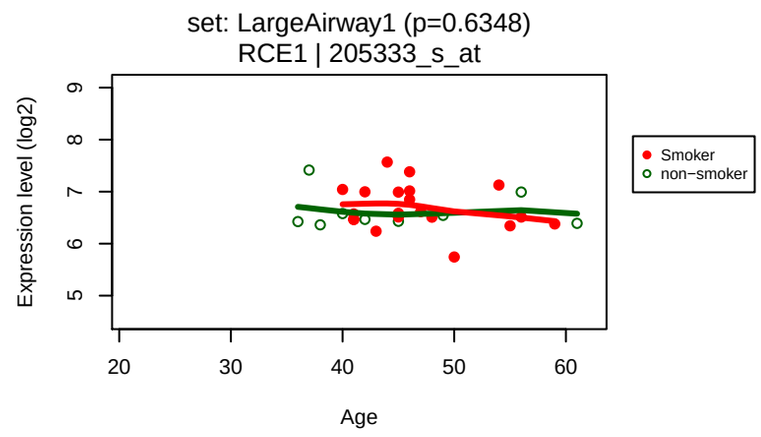

Gene expression of smokers and nonsmokers
gene: RCE1 | 205333_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3755
p=0.8093
p=0.5302
p=0.6348
p=0.2289
p=0.6971
p=0.0609
p=0.2279
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

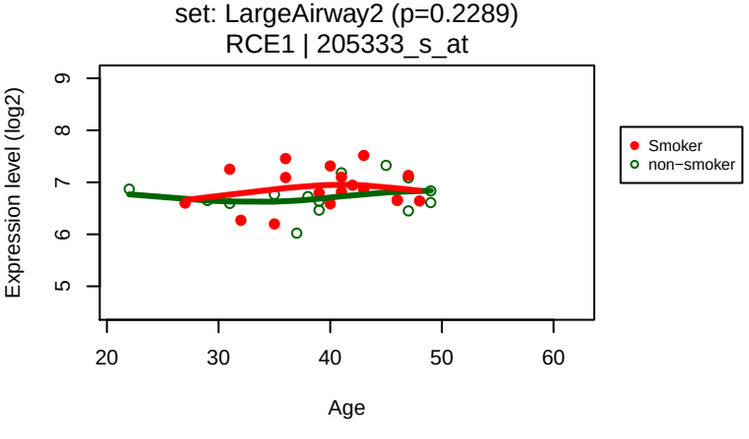

Gene expression ratio (smokers/nonsmokers)
gene: RCE1 | 205333_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3755
p=0.8093
p=0.5302
p=0.6348
p=0.2289
p=0.6971
p=0.0609
p=0.2279
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

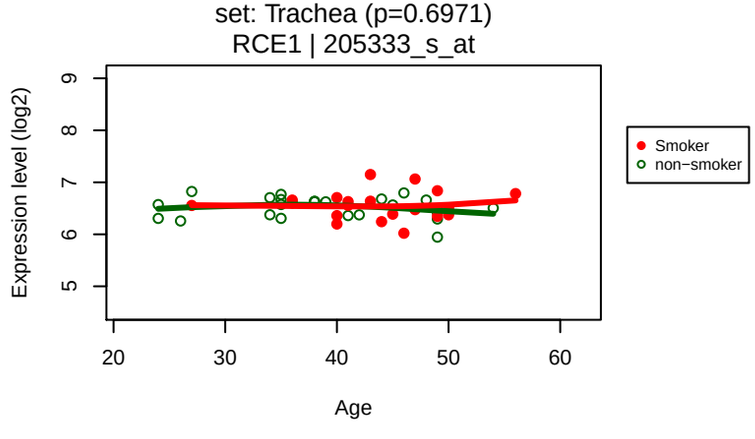

Gene expression ratio in Males
gene: RCE1 | 205333_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.561
p=0.0325
p=0.5839
p=0.3441
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RCE1 | 205333_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5359
p=0.2031
p=0.536
p=0.4843
SmallAirway2
LargeAirway2
Trachea
Macrophage2

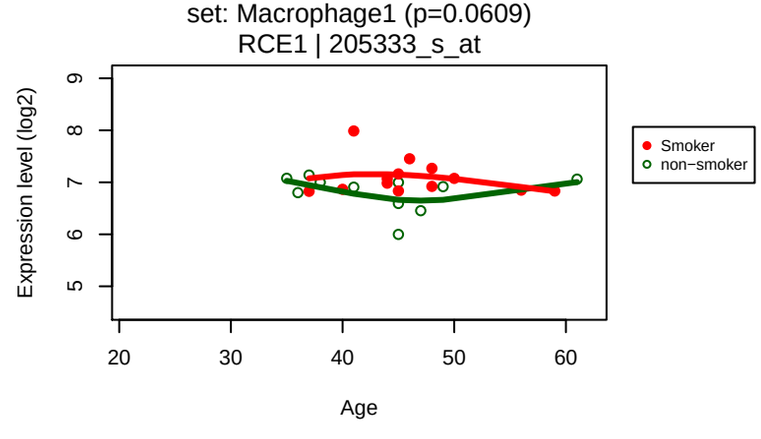

set: SmallAirway1 (p=0.3755)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker

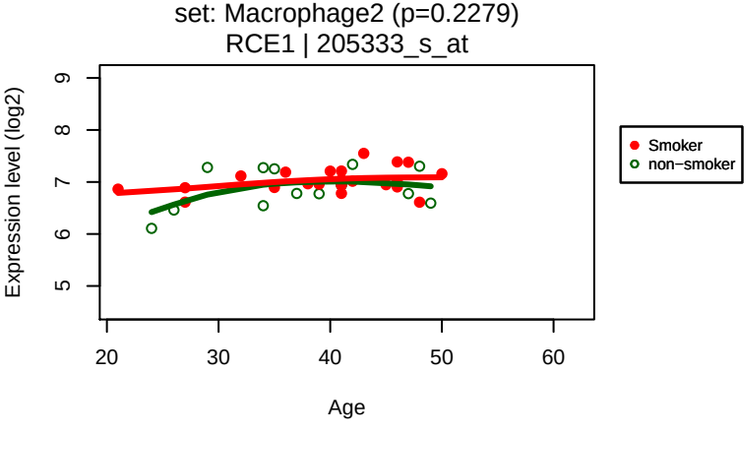

set: SmallAirway2 (p=0.8093)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker

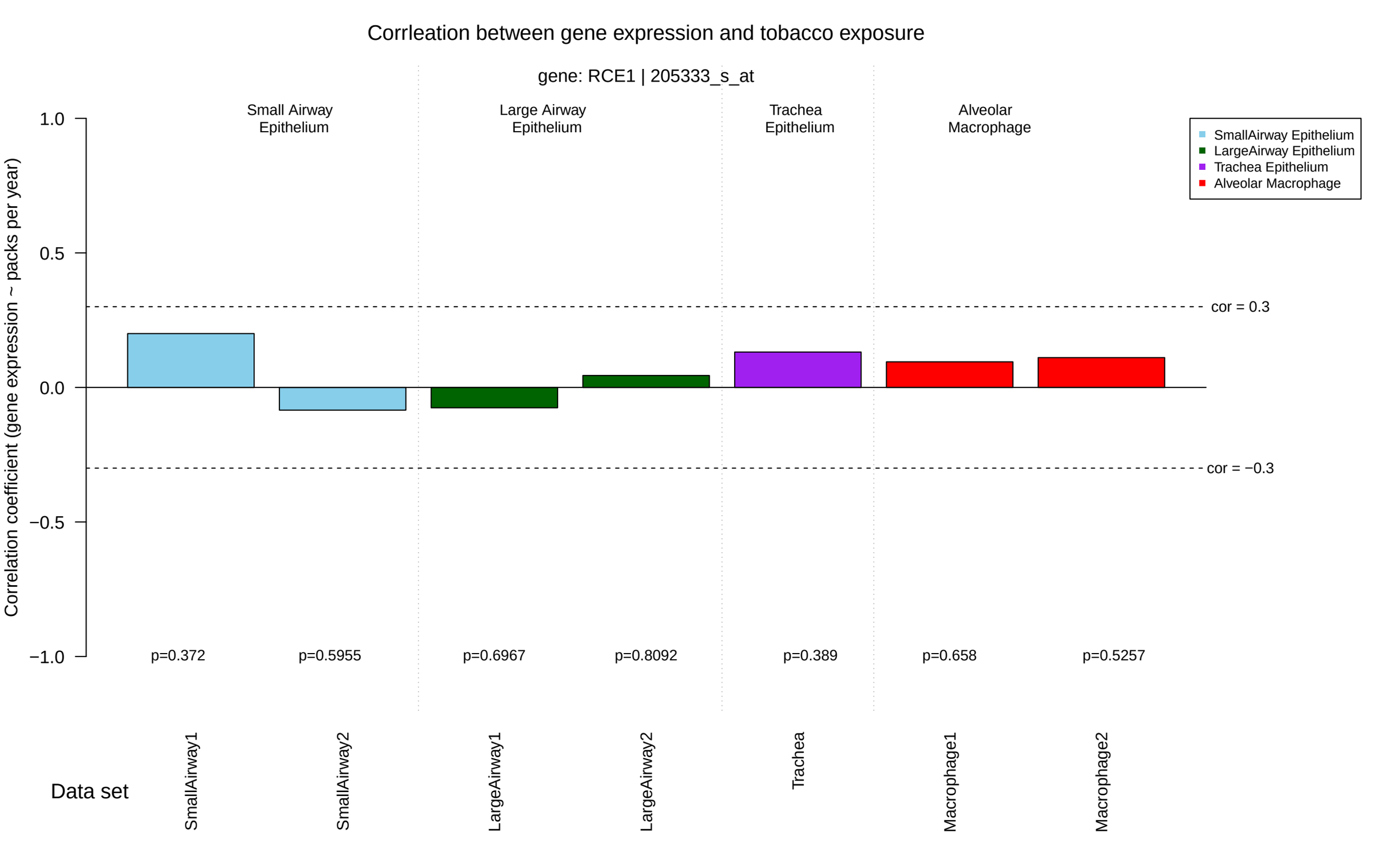

set: LargeAirway1 (p=0.6348)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2289)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6971)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0609)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2279)
RCE1 | 205333_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RCE1 | 205333_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.372
p=0.5955
p=0.6967
p=0.8092
p=0.389
p=0.658
p=0.5257
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2